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1) Problems on Gamma function

▷ Evaluate $\int_0^\infty y^6 e^{-y^6} dy$

⇒ soln

let $I = \int_0^\infty y^6 e^{-y^6} dy$

put

⇒ Prove that $\int_0^\infty \sqrt{y}\, e^{-y^2} dy \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}} dy = \frac{\pi}{2\sqrt{2}}$

⇒ soln

let $I_1 = \int_0^\infty \sqrt{y}\, e^{-y^2} dy$.

and $I_2 = \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}} dy$.

Let $I = I_1 \cdot I_2$

put $y^2 = t$ i.e. $y = \sqrt{t}$

$\therefore dy = \frac{1}{2\sqrt{t}} dt$

when $y = 0 \;;\; t = 0$

$y = \infty \;;\; t = \infty$

$\therefore I = I_1 I_2$

$= \int_0^\infty (\sqrt{t})^{\frac{1}{2}} e^{-t} \frac{1}{2\sqrt{t}} dt \int_0^\infty \frac{e^{-t}}{(\sqrt{t})^{1/2}} \frac{1}{2\sqrt{t}} dt.$

$= \frac{1}{2} \int_0^\infty e^{-t} (\sqrt{t})^{\frac{1}{2}-1} dt \; \frac{1}{2} \int_0^\infty e^{-t} (\sqrt{t})^{-\frac{1}{2}-1} dt$

$= \frac{1}{2} \int_0^\infty e^{-t} (\sqrt{t})^{-\frac{1}{2}} dt \; \frac{1}{2} \int_0^\infty e^{-t} (\sqrt{t})^{-\frac{3}{2}} dt$

$= \frac{1}{4} \int_0^\infty e^{-t} t^{-\frac{1}{4}} dt \cdot \int_0^\infty e^{-t} t^{-\frac{3}{4}} dt$

$= \frac{1}{4} \lceil -\frac{1}{4}+1 \rceil \lceil -\frac{3}{4}+1 = \frac{1}{4} \lceil \frac{3}{4} \rceil \lceil \frac{1}{4} = \frac{1}{4} \lceil \frac{1}{4} \rceil 1-\frac{1}{4}$

$= \frac{1}{4} \frac{\pi}{\sin(\frac{1}{4}\pi)}$ ..... $\left[\because \lceil n \rceil 1-n = \frac{\pi}{\sin(\pi n)}\right]$

$= \frac{1}{4} \cdot \frac{\pi}{1/\sqrt{2}} = \frac{\pi}{2\sqrt{2}\sqrt{2} \cdot \frac{1}{\sqrt{2}}} = \frac{\pi}{2\sqrt{2}}$ -- Ans.

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3) Show that $\int_0^\infty x e^{-x^8} dx \int_0^\infty x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}$

$\Rightarrow$ Soln ;- let $I = \int_0^\infty x e^{-x^8} dx \int_0^\infty x^2 e^{-x^4} dx = I_1 I_2$

where

$I_1 = \int_0^\infty x e^{-x^8} dx$ and $I_2 = \int_0^\infty x^2 e^{-x^4} dx$

put, $x^4 = t$

i.e. $x = t^{1/4}$ $\therefore dx = \frac{1}{4} t^{\frac{1}{4}-1} dt$

when $x = 0$ ; $t = 0$

$x = \infty$ ; $t = \infty$

$\therefore I_1 = \int_0^\infty x e^{-x^8} dx = \int_0^\infty x e^{-(x^4)^2} dx$

$= \int_0^\infty t^{\frac{1}{4}} e^{-t^2} \cdot \frac{1}{4} t^{\frac{1}{4}-1} dt$

$I_1 = \frac{1}{4} \int_0^\infty e^{-t^2} \cdot t^{\frac{1}{4}+\frac{1}{4}-1} \cdot dt = \frac{1}{4} \int_0^\infty e^{-t^2} \cdot t^{\frac{1}{2}-1} \cdot dt$

$I_1 = \frac{1}{8} \cdot 2 \int_0^\infty e^{-t^2} \cdot t^2 \left(\frac{1}{4}\right)^{-1} dt = \frac{1}{8} \lceil \frac{1}{4}$ ——— ②

and

$I_2 = \int_0^\infty x^2 e^{-x^4} dx = \int_0^\infty (t^{1/4})^2 \cdot e^{-t} \frac{1}{4} t^{\frac{1}{4}-1} dt$

$33 = \frac{1}{4} \int_0^\infty e^{-t} t^{\frac{3}{4}-1} dt = \frac{1}{4} \lceil \frac{3}{4}$ ——— ③

$\therefore I = I_1 I_2 = \frac{1}{8} \lceil \frac{1}{4} \cdot \frac{1}{4} \lceil \frac{3}{4}$

By eqn ①, ② and ③

$= \frac{1}{32} \lceil \frac{1}{4} \lceil 1 - \frac{1}{4}$

$= \frac{1}{32} \frac{\pi}{\sin(\pi/4)}$

$= \frac{1}{32} \cdot \frac{\pi}{1/\sqrt{2}} = \frac{\pi}{16\sqrt{2}}$

4) Evaluate $\int_0^\infty e^{-x^5} dx$

$\Rightarrow$ soln Let $I = \int_0^\infty e^{-x^5} dx$

Put $x^5 = t$

$x = t^{\frac{1}{5}} \Rightarrow dx = \frac{1}{5} t^{-4/5} dt$

| $x$ | $0$ | $\infty$ |
|---|---|---|
| $t$ | $0$ | $\infty$ |

$$= \int_0^\infty e^{-t} \frac{1}{5} t^{-\frac{4}{5}} dt = \frac{1}{5} \int_0^\infty e^{-t} t^{-\frac{4}{5}} dt$$

$$= \frac{1}{5} \int_0^\infty e^{-t} t^{\frac{1}{5}-1} dt = \frac{1}{5} \lceil \frac{1}{5} = \lceil \frac{1}{5}+1$$

$$= \lceil \frac{6}{5}$$

5) Evaluate $\int_0^\infty 7^{-4x^2} dx$

$\Rightarrow$ soln :- Let $I = \int_0^\infty 7^{-4x^2} dx$

Put $7^{-4x^2} = e^{-t}$

$\therefore \log 7^{-4x^2} = \log e^{-t} - 4x^2 (\log 7) = -t$

$x^2 = \frac{t}{4(\log 7)}$ i.e. $x = \frac{\sqrt{t}}{2\sqrt{\log 7}}$ $\therefore dx = \frac{1}{2\sqrt{\log 7}} \frac{1}{2\sqrt{t}} dt$.

when $x = 0 ; t = 0$ $x = \infty ; t = \infty$

$$I = \int_0^\infty e^{-t} \frac{1}{2\sqrt{\log 7}} \frac{1}{2\sqrt{t}} dt.$$

$$= \frac{1}{4\sqrt{\log 7}} \int_0^\infty e^{-t} t^{\frac{1}{2}-1}$$

$$dt = \frac{1}{4\sqrt{\log 7}} \sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{4\sqrt{\log 7}}$$

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6) Evaluate $\int_0^\infty x^n e^{\sqrt{a}x}\,dx$

## (2) Problems on Beta function

1) Evaluate $\int_0^2 x^4(8-x^3)^{-\frac{1}{3}}\,dx$

$\Rightarrow$ soln :-

$$I = \int_0^2 x^4 (8-x^3)^{-\frac{1}{3}}\,dx$$

put $x^3 = 8t$ i.e. $x = (8t)^{\frac{1}{3}} = 2t^{\frac{1}{3}}$

$$\therefore dx = \frac{2}{3} t^{\frac{1}{3}-1}\,dt$$

when $x=0$ ; $t=0$; $x=2$; $t=1$

$$I = \int_0^1 \left(2t^{\frac{1}{3}}\right)^4 (8-8t)^{-\frac{1}{3}} \frac{2}{3} t^{\frac{1}{3}-1}\,dt$$

$$= \int_0^1 2^4 t^{\frac{3}{4}} \{8(1-t)\}^{-\frac{1}{3}} \frac{2}{3} t^{\frac{1}{3}-1}\,dt$$

$$= \frac{2^5}{3} \int_0^1 t^{\frac{5}{3}-1} 8^{-\frac{1}{3}} (1-t)^{-\frac{1}{3}}\,dt$$

$$= \frac{2^4}{3} \int_0^1 t^{\frac{3}{5}+1} (1-t)^{\frac{2}{5}-1}\,dt$$

$$= \frac{16}{3} B\left(\frac{5}{3}, \frac{2}{3}\right)$$

$$= \frac{16 \lceil\frac{5}{3} \lceil\frac{2}{3}}{3 \lceil\frac{5}{3}+\frac{2}{3}} = \frac{16 \lceil\frac{2}{3}+1 \lceil\frac{2}{3}}{\lceil 7/3}$$

$$= \frac{16 \frac{2}{3} \lceil\frac{2}{3} \lceil\frac{2}{3}}{\frac{4}{3}\frac{1}{3} \lceil\frac{1}{3}} \quad \ldots \text{(using } \lceil n+1 = n \lceil n\text{)}$$

$$I = \frac{8 \left(\lceil\frac{2}{3}\right)^2}{\lceil\frac{1}{3}} \quad \text{Ans.}$$

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2) Evaluate $\int_0^1 \sqrt{\sqrt{x}-x}\,dx$

⟹ Soln.

Let $$I = \int_0^1 \sqrt{\sqrt{x}-x}\,dx = \int_0^1 \sqrt{\sqrt{x}(1-\sqrt{x})}\,dx$$

put $\sqrt{x} = t$ i.e. $x = t^2$ $\therefore dx = 2t\,dt$

when

$x=0$; $t=0$; $x=1$; $t=1$

$$\therefore I = \int_0^1 \sqrt{t(1-t)}\,2t\,dt$$

$$= 2\int_0^1 \sqrt{t}\,\sqrt{1-t}\,t\,dt$$

$$\therefore I = 2\int_0^1 t^{3/2}(1-t)^{1/2}\,dt$$

$$= 2B\left(\frac{3}{2}+1, \frac{1}{2}+1\right) = 2B\left(\frac{5}{2}, \frac{3}{2}\right)$$

$$= 2\,\frac{\lfloor\frac{5}{2}\,\lfloor\frac{3}{2}}{\lfloor 4} = 2\,\frac{\frac{3}{2}\frac{1}{2}\lfloor\frac{1}{2}\,\frac{1}{2}\lfloor\frac{1}{2}}{\lfloor 4}$$

$$I = \frac{3\sqrt{\pi}\sqrt{\pi}}{4\cdot 3\cdot 2\cdot 1} = \boxed{\frac{\pi}{8}}$$

3) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}} \int_0^{\frac{\pi}{2}} \sqrt{\sin\theta}\, d\theta$

⟹ soln :-

let, $I = \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}} \int_0^{\frac{\pi}{2}} \sqrt{\sin\theta}\, d\theta$

$= \int_0^{\frac{\pi}{2}} (\sin\theta)^{-\frac{1}{2}} (\cos\theta)^0 d\theta \int_0^{\pi/2} (\sin\theta)^{1/2} (\cos\theta)^0 d\theta$

$= \frac{1}{2} B\left(\frac{-\frac{1}{2}+1}{2}, \frac{0+1}{2}\right) \cdot \frac{1}{2} B\left(\frac{\frac{1}{2}+1}{2}, \frac{0+1}{2}\right)$

using $\int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$I = \frac{1}{2} B\left(\frac{1}{4}, \frac{1}{2}\right) \cdot \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{2}\right)$

$= \frac{1}{4} \cdot \frac{\lceil\frac{1}{4}\lceil\frac{1}{2}}{\lceil\frac{1}{4}+\frac{1}{2}} \cdot \frac{\lceil\frac{3}{4}\lceil\frac{1}{2}}{\lceil\frac{3}{4}+\frac{1}{2}}$

$= \frac{1}{4} \frac{\lceil\frac{1}{4}\sqrt{\pi}\,\lceil\frac{3}{4}\sqrt{\pi}}{\lceil\frac{3}{4}\lceil\frac{5}{4}} = \frac{1}{4} \frac{\lceil\frac{1}{4}\pi}{\lceil\frac{5}{4}} = \frac{1}{4} \frac{\lceil\frac{1}{4}\pi}{\frac{1}{4}\lceil\frac{1}{4}}$

---- $[\because \lceil n+1 = n\lceil n]$

$\boxed{I = \pi}$

4) Evaluate $\int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta$

$\Rightarrow$ soln

$$I = \int_0^{\frac{\pi}{2}} \sqrt{\cot\theta}\, d\theta = \int_0^{\frac{\pi}{2}} (\cos\theta)^{\frac{1}{2}} \cdot (\sin\theta)^{\frac{-1}{2}}\, d\theta$$

$$= \frac{1}{2} B\left(\frac{\frac{-1}{2}+1}{2}, \frac{\frac{1}{2}+1}{2}\right) = \frac{1}{2} B\left(\frac{1}{4}, \frac{3}{4}\right)$$

$$= \frac{1}{2} \frac{\left|\frac{1}{4}\right| \left|\frac{3}{4}\right|}{\left|1} = \frac{1}{2} \left|\frac{1}{4}\right| \left|1-\frac{1}{4}\right.$$

$$= \frac{1}{2} \frac{\pi}{\sin\left(\frac{\pi}{4}\right)} \quad \cdots \left[\because \lceil n \lceil 1-n = \frac{\pi}{\sin(n\pi)}\right]$$

$$I = \frac{1\pi}{2\frac{1}{\sqrt{2}}} = \frac{\pi}{\sqrt{2}}$$

5) Evaluate $\int_0^{\pi/6} \cos^6(3\theta) \sin^2(6\theta)\, d\theta$

$\Rightarrow$ soln;

let, $I = \int_0^{\frac{\pi}{6}} \cos^3(3\theta) \sin^2(6\theta) d\theta$

put,

$3\theta = t \Rightarrow d\theta = \frac{1}{3} dt$

when, $\theta = 0, t = 0 \qquad \theta = \frac{\pi}{6}, t = \frac{\pi}{2}$

$$I = \int_0^{\pi/2} \cos^3 t \cdot \sin^2(2t) \frac{1}{3} dt$$

$$= \frac{1}{3}\int_0^{\frac{\pi}{2}} \cos^3 t\,(2\sin t\cdot\cos t)^2\,dt$$

$$= \frac{1}{3}\int_0^{\pi/2} \cos^3 t\, 4\sin^2 t\cos^2 t\,dt$$

$$= \frac{4}{3}\int_0^{\pi/2} \sin^2 t\cos^5 t\,dt$$

$$= \frac{41}{32} B\left(\frac{2+1}{2}, \frac{5+1}{2}\right) = \frac{2}{3} B\left(\frac{3}{2}, 3\right)$$

$$= \frac{2}{3}\frac{\sqrt{\frac{3}{2}}\sqrt{3}}{\sqrt{\frac{3}{2}+3}} = \frac{2}{3}\frac{\frac{1}{2}\sqrt{\frac{1}{2}}\,2!}{\sqrt{\frac{9}{2}}}$$

$$= \frac{2}{3}\frac{\sqrt{\pi}}{\frac{7}{5}\frac{5}{3}\frac{3}{2}\frac{1}{2}\sqrt{\pi}}$$

$$I = \frac{32}{315}$$

6) Evaluate $\int_0^\infty \left(\frac{x}{1+x^2}\right)^4 dx$

$\Longrightarrow$ soln

let, $I = \int_0^\infty \left(\frac{x}{1+x^2}\right)^4 dx$

Put $x^2 = t$ i.e $x = \sqrt{t}$

$$dx = \frac{1}{2\sqrt{t}}\,dt$$

when, $x = 0$, $t = 0$

$x = \infty$ $t = \infty$

$$\therefore I = \int_0^\infty \left(\frac{\sqrt{t}}{1+t}\right)^4 \frac{1}{2\sqrt{t}}\,dt = \frac{1}{2}\int_0^\infty \frac{t^{\frac{3}{2}}}{(1+t)^4}\,dt$$

$$= \frac{1}{2}\int_0^\infty \frac{t^{\frac{5}{2}-1}}{(1+t)^{5/2+3/2}}\,dt = \frac{1}{2} B\left(\frac{5}{2}, \frac{3}{2}\right)$$

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$$= \frac{1}{2}\frac{\Gamma\frac{5}{2}\Gamma\frac{3}{2}}{\Gamma\frac{8}{2}} = \frac{1}{2}\frac{\frac{3}{2}\frac{1}{2}\sqrt{\pi}\,\frac{1}{2}\sqrt{\pi}}{\sqrt{4}}$$

$$I = \frac{3\pi}{163} = \frac{\pi}{32}$$

$$\boxed{I = \frac{\pi}{32}}$$

7) Evaluate $\int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}}\,dx$

→ soln.

Let,

$$I = \int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}}\,dx = \int_0^\infty \frac{x^8 - x^{14}}{(1+x)^{24}}\,dx$$

$$= \int_0^\infty \frac{x^8}{(1+x)^{24}}\,dx - \int_0^\infty \frac{x^{14}}{(1+x)^{24}}\,dx$$

$$= \int_0^\infty \frac{x^{9-1}}{(1+x)^{9+15}}\,dx - \int_0^\infty \frac{x^{15-1}}{(1+x)^{15+9}}\,dx$$

$$\therefore I = B(9,15) - B(15,9)$$

$$I = B(15,9) - B(15,9) = 0$$

8) Evaluate $\int_3^7 \sqrt[4]{(x-3)(7-x)}\,dx$

$\Rightarrow$ let, $I = \int_3^7 \sqrt[4]{(x-3)(7-x)}\,dx$

put $x-3 = (7-3)t$ i.e. $x = 3+4t$

$\therefore dx = 4dt$

when $x=3$ ; $t=0$

$x=7$ ; $t=1$

$$I = \int_0^1 \sqrt[4]{4t(7-(3+4t))}\,4\,dt$$

$$= 4\int_0^1 (4t)^{\frac{1}{4}}(4-4t)^{\frac{1}{4}}\,dt$$

$$= 4\int_0^1 4^{\frac{1}{4}} t^{\frac{1}{4}} (4(1-t))^{\frac{1}{4}}\,dt$$

$$= 4\cdot 4^{\frac{1}{4}} \int_0^1 t^{\frac{1}{4}} 4^{\frac{1}{4}} (1-t)^{\frac{1}{4}}\,dt$$

$$= 4^{\frac{1}{4}+\frac{1}{4}+1} \int_0^1 t^{\frac{1}{4}}(1-t)^{\frac{1}{4}}\,dt$$

$$= 4^{\frac{3}{2}} \cdot B\left(\tfrac{1}{4}+1, \tfrac{1}{4}+1\right)$$

$$= \frac{4\sqrt{4}\ \left\lceil \frac{1}{4}+1 \right\rceil \left\lceil \frac{1}{4}+1 \right\rceil}{\left\lceil \frac{1}{4}+1+\frac{1}{4}+1 \right\rceil} = 8\,\frac{\frac{1}{4}\left\lceil\frac{1}{4}\ \frac{1}{4}\right\lceil\frac{1}{4}}{\left\lceil\frac{5}{2}\right.}$$

$$I = \frac{1}{2}\,\frac{\left(\left\lceil\frac{1}{4}\right.\right)^2}{\frac{3}{2}\,\frac{1}{2}\sqrt{\pi}} = \frac{2}{3}\,\frac{\left(\left\lceil\frac{1}{4}\right.\right)^2}{\sqrt{\pi}}$$

$$\boxed{I = \frac{2\left(\left\lceil\frac{1}{4}\right.\right)^2}{3\sqrt{\pi}}}$$

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11) If $\beta(n,3) = \frac{1}{105}$ and n is a positive integer than find n.

⟹ soln

$$\beta(n,3) = \frac{1}{105} \Rightarrow \frac{\lfloor n \lfloor 3}{\lfloor n+3} = \frac{1}{105}$$

$$\frac{\lfloor n \; 2!}{(n+2)(n+1)n \lfloor n} = \frac{1}{105}$$

$$\Rightarrow 210 = n(n+1)(n+2)$$

$$\therefore n^3 + 3n^2 + 2n - 210 = 0$$

$$\therefore \boxed{n = 5} \quad \because n \text{ is a positive integer}$$

3) Problems on Differentiation under Integral sign (DUIS)

1) show that $\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx = \frac{\pi}{2}\log(1+a)$

⟹ soln :-

$$I(a) = \int_0^\infty \frac{\partial}{\partial a}\left[\frac{\tan^{-1}(ax)}{x(1+x^2)}\right]dx$$

$$= \int_0^\infty \frac{1}{x(1+x^2)}\,\frac{x}{1+a^2x^2}\,dx$$

$$= \int_0^\infty \left[\frac{1}{1+x^2}\,\frac{1}{1+a^2x^2}\right]dx$$

$$\frac{dI}{da} = \frac{1}{1-a^2}\int_0^\infty \left(\frac{1}{1+x^2} - \frac{a^2}{1+a^2x^2}\right)dx$$

$$= \frac{1}{1-a^2}\left[\tan^{-1}x - a\tan^{-1}(ax)\right]_0^\infty$$

$$= \frac{1}{1-a^2}\left[\frac{\pi}{2} - a\frac{\pi}{2}\right] = \frac{\pi}{2}\,\frac{(1-a)}{1-a^2} = \frac{\pi}{2}\,\frac{1}{1+a}$$

Integrating both sides, we get

$$\int dI = \frac{\pi}{2}\int \frac{1}{1+a}\,da + c$$

$$I(a) = \frac{\pi}{2}\log(1+a) + c$$

To find value of c, we put $a = 0$,

we get $I(0) = \frac{\pi}{2}\log(1+0) + c$

from eqn(1), we have

$$0 = 0 + c$$

From eqn(2), we get

$$I(a) = \frac{\pi}{2}\log(1+a)$$

$$\boxed{\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx = \frac{\pi}{2}\log(1+a)}$$

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2) evaluate $\int_0^1 \frac{x^a - 1}{\log x}\, dx$

$\Longrightarrow$ soln :-

Let $I(a) = \int_0^1 \frac{x^a - 1}{\log x}\, dx$

using DUIS with 'a' as parameter, we have

$$\frac{dI}{da} = \int_0^1 \frac{\partial}{\partial a}\left[\frac{x^a - 1}{\log x}\right] dx$$

$$= \int_0^1 \frac{1}{\log x} \frac{\partial}{\partial a}\left[x^a - 1\right] dx$$

$$= \int_0^1 \frac{1}{\log x}\left[x^a \cdot \log x\right] dx$$

$$= \int_0^1 x^a\, dx = \left[\frac{x^{a+1}}{a+1}\right]_0^1 = \frac{1}{a+1} - 0$$

$$\therefore \frac{dI}{da} = \frac{1}{a+1} \quad \therefore dI = \frac{1}{a+1}\, da$$

$$\therefore I(a) = \log(a+1) + c \quad \text{---} (2)$$

put $a = 0$ in above eqn, we get

$$I(0) = \log(0+1) + c$$

$$0 = 0 + c \quad \therefore c = 0$$

from eqn (2), we have

$$I(a) = \log(a+1)$$

$$\therefore \int_0^1 \frac{x^a - 1}{\log x}\, dx = \log(a+1)$$

3) Prove the $\int_0^\infty \frac{\log(1+ax^2)}{x^2}\,dx = \pi\sqrt{a}$

$\Rightarrow$ let $I(a) = \int_0^\infty \frac{\log(1+ax^2)}{x^2}\,dx$ ---(1)

By leibnitz's rule of DUIS

we get $\frac{dI}{da} = \int_0^\infty \frac{\partial}{\partial a}\left(\frac{\log(1+ax^2)}{x^2}\right)dx$

$= \int_0^\infty \frac{1}{x^2}\,\frac{1}{(1+ax^2)}\,x^2\,dx$

$= \int_0^\infty \frac{1}{(1+ax^2)}\,dx = \int_0^\infty \frac{1}{(1+(\sqrt{a}x)^2)}\,dx$

$= \left[\frac{\tan^{-1}(\sqrt{a}x)}{\sqrt{a}}\right]_0^\infty$

$= \frac{1}{\sqrt{a}}[\tan^{-1}(\infty) - \tan^{-1}(0)] = \frac{1}{\sqrt{a}}\left(\frac{\pi}{2} - 0\right)$

$\therefore \frac{dI}{da} = \frac{\pi}{2\sqrt{a}}$

$\therefore dI = \frac{\pi}{2\sqrt{a}}\,da$

Intergating, we get

$I(a) = \int \frac{\pi}{2\sqrt{a}}\,da + c = \pi\sqrt{a} + c.$

To find c put $a = 0$

$\therefore I(0) = \pi\sqrt{0} + c$

$\therefore 0 = 0 + c$ --- [$\because I(0) = 0$ by equation (1)]

$\therefore c = 0$

$\therefore I(a) = \pi\sqrt{a}$

i.e. $\int_0^\infty \frac{\log(1+ax^2)}{x^2}\,dx = \boxed{\pi\sqrt{a}}$

4) Evaluate $\int_0^{\pi} \frac{\log(1+a\cos x)}{\cos x}\,dx = \pi \sin^{-1} a$ for $0<a<1$

$\Rightarrow$ By the DUIS rule we get

$$\frac{dI}{da} = \int_0^{\pi} \frac{\partial}{\partial a}\left(\frac{\log(1+a\cos x)}{\cos x}\right)dx$$

$$= \int_0^{\pi} \frac{1}{\cos x} \cdot \frac{1}{(1+a\cos x)} \cdot \cos x\,dx$$

$$= \int_0^{\pi} \frac{1}{(1+a\cos x)}\,dx$$

put $\tan\frac{x}{2} = t$

and $\cos x = \frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}$

$$x = 2\tan^{-1}(t)$$

$$\therefore \cos x = \frac{1-t^2}{1+t^2}$$

$$\therefore dx = \frac{2}{1+t^2}\,dt$$

when $x=0 \;;\; t=0$
$x=\pi \;;\; t=\infty$

$$\therefore \frac{dI}{da} = \int_0^{\infty} \frac{1}{\left(1+a\left(\frac{1-t^2}{1+t^2}\right)\right)} \frac{2}{(1+t^2)}\,dt$$

$$\frac{dI}{da} = \int_0^\infty \frac{1}{\frac{(1+t^2)+a(1-t^2)}{(1+t^2)}} \cdot \frac{2}{(1+t^2)}\,dt.$$

$$= \int_0^\infty \frac{2}{(1+t^2+a-at^2)}\,dt$$

$$= 2\int_0^\infty \frac{1}{(1+a)+(1-a)t^2}\,dt$$

$$= \frac{2}{(1+a)} \int_0^\infty \frac{1}{\left(1+\left(\frac{1-a}{1+a}\right)t^2\right)}\,dt$$

$$= \frac{2}{(1+a)} \left[\frac{\tan^{-1}\left(\sqrt{\frac{1-a}{1+a}}\,t\right)}{\sqrt{\frac{1-a}{1+a}}}\right]_0^\infty$$

$$= \frac{2}{(1+a)} \left[\frac{\tan^{-1}(\infty) - \tan^{-1}(0)}{\sqrt{1-a}/\sqrt{1+a}}\right]$$

$$= \frac{2}{\sqrt{1+a}\,\sqrt{1-a}} \left(\frac{\pi}{2} - 0\right)$$

$$\frac{dI}{da} = \frac{\pi}{\sqrt{1-a^2}}$$

$$\frac{dI}{da} = \frac{\pi}{\sqrt{1-a^2}}$$

$$dI = \frac{\pi}{\sqrt{1-a^2}}\,da \quad \text{Integrating, we get.}$$

$$\therefore I(a) = \int \frac{\pi}{\sqrt{1-a^2}}\,da + c = \pi \sin^{-1}(a) + c.$$

To find c put $a=0$

$\therefore I(0) = \pi \sin^{-1}(0) + c$

$0 = 0 + c$

$c = 0$ ... $(\because I(0) = 0$ by equation (1)$]$

$$\boxed{I(a) = \pi . \sin^{-1}(a)}$$

5) Evaluate $\int_0^{\pi/2} \frac{dx}{1+a\cos^2 x}$ and hence deduce that

$$\int_0^{\pi/2} \frac{\cos^2 x}{(3+\cos^2 x)^2}\,dx = \frac{\pi\sqrt{3}}{96}$$

$\Rightarrow$ let. $I = \int_0^{\pi/2} \frac{dx}{(1+a\cos^2 x)}$

$$= \int_0^{\pi/2} \frac{\sec^2 x\,dx}{(\sec^2 x + a)} \quad \text{--- [dividing by } \cos^2 x]$$

$$= \int_0^{\pi/2} \frac{\sec^2 x.dx}{(1+a)+\tan^2 x}$$

put $\tan x = t$

$\therefore \sec^2 x\,dx = dt$, whe $x = 0$; $t = 0$

$x = \pi/2$ $t = \infty$

$$\therefore I = \int_0^{\infty} \frac{dt}{(1+a)+t^2} = \int_0^{\infty} \frac{dt}{(\sqrt{1+a})^2+t^2}$$

$$= \frac{1}{\sqrt{1+a}}\left[\tan^{-1}\left(\frac{t}{\sqrt{1+a}}\right)\right]_0^{\infty}$$

$$= \frac{1}{\sqrt{1+a}}\left(\frac{\pi}{2}-0\right)$$

$$\therefore I = \frac{\pi}{2\sqrt{1+a}} \quad \text{i.e } I(a)$$

$$= \int_0^{\pi/2} \frac{dx}{(1+a\cos^2 x)} = \frac{\pi}{2\sqrt{1+a}} \quad \text{--- (1)}$$

By DUIS rule we get

$$\frac{dI}{da} = \int_0^{\pi/2} \frac{\partial}{\partial a}\left(\frac{1}{1+a\cos^2 x}\right) dx$$

$$\therefore \frac{dI}{da} = \int_0^{\pi/2} \frac{-1}{(1+a\cos^2 x)^2} \cos^2 x \, dx$$

By equation (1)

$$\frac{dI}{dx} = \frac{d}{da}\left(\frac{\pi}{2\sqrt{1+a}}\right)$$

$$= \frac{\pi}{2}\left(\frac{-1}{2}\right)(1+a)^{-3/2} = \frac{-\pi}{4(1+a)^{3/2}}$$

$$\therefore \int_0^{\pi/2} \frac{-\cos^2 x}{(1+a\cos^2 x)^2} dx = \frac{\pi}{4(1+a)^{3/2}}$$

put $a = \frac{1}{3}$

$$\int_0^{\pi/2} \frac{\cos^2 x}{\left(1+\frac{1}{3}\cos^2 x\right)^2} dx = \frac{\pi}{4\left(1+\frac{1}{3}\right)^{3/2}}$$

$$\int_0^{\pi/2} \frac{9\cos^2 x}{(3+\cos^2 x)^2} dx = \frac{\pi}{4\left(\frac{4}{3}\right)^{3/2}} = \frac{3\sqrt{2}\pi}{32}$$

$$\int_0^{\pi/2} \frac{\cos^2 x}{(3+\cos^2 x)^2} dx = \frac{\pi\sqrt{3}}{96}$$

$$= \boxed{\frac{\pi\sqrt{3}}{96}}$$